The Quantum-to-Classical Transition: Environmental Decoherence and Gravitationally Induced Collapse

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Abstract

We formalize the emergence of classical macroscopic definiteness from unitary quantum superpositions by synthesizing open-system environmental decoherence with gravitationally induced objective collapse. Using the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation and Caldeira–Leggett thermal photon scattering alongside the Penrose–Diósi gravitational self-energy functional and stochastic Schrödinger dynamics, we establish a unified dimensionless diagnostic framework. Specifically, we introduce the Decoherence Action (Ω), Timeline Integrity (Φ_τ), the Holographic Coherence Floor (Φ_{floor}), Surviving Coherent Cycles ($\mathcal{N}_{\text{eff}}$), and the Buckingham-Π Gravity-to-Thermal Crossover Ratio ($\mathcal{R}_{\text{DP}}$) to map mesoscopic quantum-to-classical boundaries across relativistic worldlines.

Keywords: Quantum-to-Classical Transition, Environmental Decoherence, GKSL Master Equation, Caldeira–Leggett Scattering, Penrose–Diósi Collapse, Holographic Coherence Floor

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1 Open-System Quantum Evolution and Environmental Decoherence (Γ_{th})

In closed quantum mechanics, a pure state $|\psi\rangle$ evolves unitarily under the Schrödinger equation, preserving phase coherence indefinitely. To model how a quantum system of characteristic radius L and effective mass M_{eff} loses phase coherence when coupled to an ambient thermal bath at local temperature T_{eff}, we trace out the environmental degrees of freedom ($\rho_S = \text{Tr}_E[\rho_{\text{tot}}]$).

1.1 Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) Master Equation

Under the Born–Markov approximation, the reduced density matrix ρ_S evolves along the system's proper worldline (τ_{pre}) according to:

$$\frac{d\rho_S}{d\tau_{\text{pre}}} = -\frac{i}{\hbar}[H_S, \rho_S] + \sum_k \gamma_k \left(L_k \rho_S L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho_S\} \right), \tag{1}$$

where L_k denote Lindblad jump operators representing momentum-transfer collisions with ambient thermal quanta and $\gamma_k \geq 0$ are non-negative dissipation rates [1, 2].

1.2 Feynman–Vernon Influence Functional and Caldeira–Leggett Scattering

In the position representation $\rho_S(\mathbf{x}, \mathbf{x}', \tau_{\text{pre}}) = \langle \mathbf{x}|\rho_S|\mathbf{x}'\rangle$, integrating out a bath of blackbody photons or harmonic oscillators produces an exponential off-diagonal damping kernel:

$$\frac{\partial \rho_S(\mathbf{x}, \mathbf{x}', \tau_{\text{pre}})}{\partial \tau_{\text{pre}}} = -\frac{i}{\hbar}\langle \mathbf{x}|[H_S, \rho_S]|\mathbf{x}'\rangle - \Lambda(\Delta x)\, \rho_S(\mathbf{x}, \mathbf{x}', \tau_{\text{pre}}). \tag{2}$$

For a spatial superposition separated by $\Delta x \sim L$ immersed in a thermal radiation bath with reduced thermal wavelength $\lambda\!\!\!^{-}_{\text{th}} = \hbar c/(k_B T_{\text{eff}})$, dimensional photon-scattering scaling yields the **Caldeira–Leggett thermal decoherence rate** [3, 4]:

$$\Gamma_{\text{th}} = \frac{L^2 (k_B T_{\text{eff}})^3}{(\hbar c)^2 \hbar} = \frac{c\, L^2}{\lambda\!\!\!^{-3}_{\text{th}}}. \tag{3}$$

2 Gravitationally Induced Objective Collapse (Γ_{grav})

Even at absolute zero ($T_{\text{eff}} = 0$ K) in an ideal vacuum, the **Penrose–Diósi hypothesis** proposes that placing a massive object into a spatial superposition of two distinct mass distributions $\rho_1(\mathbf{x})$ and $\rho_2(\mathbf{x})$ generates an irreducible spacetime curvature tension [5, 6].

2.1 Regularized Gravitational Self-Energy Functional

The gravitational interaction energy difference ΔE_{grav} between two superposed mass configurations is governed by the Newtonian Green's function of the Laplacian:

$$\Delta E_{\text{grav}} = -G \iint \frac{[\rho_1(\mathbf{x}) - \rho_2(\mathbf{x})]\,[\rho_1(\mathbf{y}) - \rho_2(\mathbf{y})]}{|\mathbf{x} - \mathbf{y}|}\, d^3x\, d^3y. \tag{4}$$

For a localized mass M_{eff} displaced across its characteristic radius L, this self-energy difference scales as:

$$\Delta E_{\text{grav}} \sim \frac{G M_{\text{eff}}^2}{L} = \left(\frac{M_{\text{eff}}}{M_P}\right)^2 \frac{\hbar c}{L}, \tag{5}$$

where $M_P = \sqrt{\hbar c/G}$ is the Planck mass.

2.2 Stochastic Schrödinger Evolution and Collapse Rate

Introducing continuous non-linear stochastic localization driven by Wiener noise dW_t—preserving the Born rule via Girsanov's theorem without superluminal signaling—yields a pure-state collapse timescale $\tau_{\text{DP}} = \hbar/\Delta E_{\text{grav}}$, defining the **Penrose–Diósi gravitational dephasing rate**:

$$\Gamma_{\text{grav}} = \frac{\Delta E_{\text{grav}}}{\hbar} = \left(\frac{M_{\text{eff}}}{M_P}\right)^2 \frac{c}{L}. \tag{6}$$

3 Dimensionless Diagnostic Synthesis

To evaluate phase survival and regime dominance across relativistic worldlines without coordinate ambiguity, we synthesize the competing decoherence rates into dimensionless invariants.

3.1 Decoherence Action (Ω) and ZAMO Dilation Factor (ρ_τ)

Converting distant observer stopwatch time t_{obs} into proper worldline time $t_{\text{pre}} = t_{\text{obs}}/\rho_\tau$ via the **ZAMO Dilation Factor (ρ_τ)**, the accumulated environmental and gravitational phase-scattering depth is defined by the **Decoherence Action**:

$$\Omega = \int_0^{t_{\text{pre}}} (\Gamma_{\text{th}} + \Gamma_{\text{grav}})\, d\tau = (\Gamma_{\text{th}} + \Gamma_{\text{grav}})\, \frac{t_{\text{obs}}}{\rho_\tau}. \tag{7}$$

3.2 Timeline Integrity (Φ_τ) and Holographic Coherence Floor (Φ_{floor})

Off-diagonal density-matrix coherence decays exponentially with Ω, bounded from below by the inverse Bekenstein–Hawking Hilbert-space dimension $\Phi_{\text{floor}} = (1 + \pi \cdot 10^{2\zeta})^{-1}$ (where $\zeta = \log_{10}(L/\ell_P)$) [9]:

$$\Phi_\tau(\Omega) = \Phi_{\text{floor}} + (1 - \Phi_{\text{floor}})\, e^{-\Omega}. \tag{8}$$

- **Unitary Quantum Regime ($\Omega \ll 1$):** $\Phi_\tau \approx 1.0000$, preserving coherent interference.
- **Classical Thermalized Equilibrium ($\Omega \gg 1$):** $\Phi_\tau \to \Phi_{\text{floor}} > 0$, enforcing strict quantum unitarity down to the Planck scale ($\Phi_{\text{floor}} = (1+\pi)^{-1} \approx 0.2414$ at $L = \ell_P$) while reproducing classical decoherence ($\Phi_{\text{floor}} \sim 10^{-70}$) at macroscopic scales ($L = 1$ m).

3.3 Gravity-to-Thermal Crossover Ratio ($\mathcal{R}_{\text{DP}}$)

Constructing the dimensionless Buckingham-Π ratio between gravitational self-energy collapse and thermal scattering cancels time units completely:

$$\mathcal{R}_{\text{DP}} \equiv \frac{\Gamma_{\text{grav}}}{\Gamma_{\text{th}}} = \left(\frac{M_{\text{eff}}}{M_P}\right)^2 \left(\frac{\lambda\!\!\!^{-}_{\text{th}}}{L}\right)^3. \tag{9}$$

- **Thermal Dominance ($\mathcal{R}_{\text{DP}} < 1$):** Environmental heat destroys phase coherence prior to gravitational self-energy collapse [7].
- **Exact Mesoscopic Crossover ($\mathcal{R}_{\text{DP}} = 1$):** Occurs at $M_{\text{eff}}/M_P = (L/\lambda\!\!\!^{-}_{\text{th}})^{3/2}$, isolating the critical mass-temperature boundary targeted by cryogenic levitated-nanoparticle experiments.
- **Curvature Dominance ($\mathcal{R}_{\text{DP}} > 1$):** Gravitational self-energy enforces classical spatial localization even in ultracold vacuum traps.

3.4 Surviving Coherent Cycles ($\mathcal{N}_{\text{eff}}$) and Semiclassical Guard (Ξ_{QG})

Weighting total proper wave oscillations $\mathcal{N}_\phi = f_{\text{local}}\, t_{\text{pre}}$ by Φ_τ yields $\mathcal{N}_{\text{eff}} = \mathcal{N}_\phi \cdot \Phi_\tau$, quantifying the exact number of usable interference cycles prior to classicalization [8]. Simultaneously, the L_∞ semiclassical monitor Ξ_{QG} tracks ℓ_P/L and T_{eff}/T_P to ensure localization scales and bath temperatures remain strictly within semiclassical bounds.

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